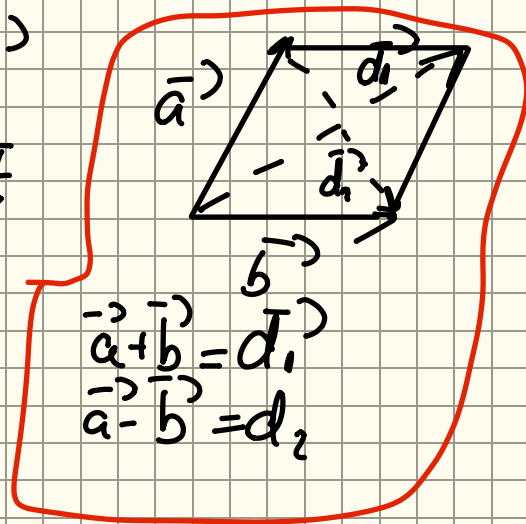


$$1.7) \vec{a} = 5\vec{p} - 3\vec{q}, \vec{b} = \vec{p} + 2\vec{q}$$

$$\|\vec{p}\| = 3, \quad \alpha = \angle(\vec{p}, \vec{q}) = \frac{\pi}{3}$$

$$|\vec{q}| = 2$$



$$a) \vec{a} + \vec{b} = 6\vec{p} - \vec{q}$$

$$\vec{a} - \vec{b} = 4\vec{p} - 5\vec{q}$$

$$\vec{p} \cdot \vec{q} = |\vec{p}| \cdot |\vec{q}| \cdot \cos \alpha = 3 \cdot 2 \cdot \frac{1}{2} = 3$$

$$|\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = (6\vec{p} - \vec{q}) \cdot (6\vec{p} - \vec{q})$$

$$= 36|\vec{p}|^2 - 12\vec{p} \cdot \vec{q} + |\vec{q}|^2 = 36 \cdot 9 - 12 \cdot 3 + 4$$

$$= 288 + 1 = 289 \Rightarrow \vec{a} + \vec{b} = \sqrt{289}$$

$$|\vec{a} - \vec{b}|^2 = 16|\vec{p}|^2 - 4\vec{p} \cdot \vec{q} + 25|\vec{q}|^2 = 16 \cdot 9 - 12 \cdot 3 + 100$$

$$= 144 - 36 + 100 = 208$$

$$\Rightarrow |\vec{a} - \vec{b}| = \sqrt{208}$$

$$b) \vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta$$

$$|\vec{a}|^2 = \vec{a} \cdot \vec{a} = 25|\vec{p}|^2 - 30\vec{p} \cdot \vec{q} + 9|\vec{q}|^2$$

$$= 225 - 90 + 36 = 135 + 36 = 171 \Rightarrow |\vec{a}| = \sqrt{171}$$

$$|\vec{b}|^2 = \vec{b} \cdot \vec{b} = |\vec{p}|^2 + 4\vec{p} \cdot \vec{q} + 4|\vec{q}|^2 = 9 + 12 + 16$$

$$= 35$$

$$\Rightarrow |\vec{b}| = \sqrt{35}, \quad \theta = \angle(\vec{d}_1, \vec{d}_2)$$

$$\Rightarrow \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a} + \vec{b}| \cdot |\vec{a} - \vec{b}| \cdot \cos \theta$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = (6\vec{p} - \vec{q}) \cdot (4\vec{p} - 5\vec{q}) =$$

$$= 24|\vec{p}|^2 - 36\vec{p} \cdot \vec{g} - 4\vec{p} \cdot \vec{g} + 5 \cdot |\vec{g}|^2$$

$$= 24 \cdot 9 - 36 \cdot 3 + 5 \cdot 4 = 216 - 108 + 20 = 128$$

$$\Rightarrow \cos \theta = \frac{128}{\sqrt{171} \cdot \sqrt{292}} \Rightarrow \theta = \arccos\left(\frac{128}{\sqrt{171} \cdot \sqrt{292}}\right)$$

$$c) |\vec{a} \times \vec{b}| = |(5\vec{p} - 3\vec{g}) \times (\vec{p} + 2\vec{g})|$$

$$= |5\vec{p} \times \vec{p} + 5 \cdot 2(\vec{p} \times \vec{g}) - 3(\vec{g} \times \vec{p}) - 6(\vec{g} \times \vec{g})|$$

$$= 10 \cdot 3\sqrt{3} + 3 \cdot 3\sqrt{3} = 39\sqrt{3}$$

$$|\vec{p} \times \vec{g}| = |\vec{p}| \cdot |\vec{g}| \cdot \sin \frac{\pi}{3} = 3 \cdot 7 \cdot \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

$$1.8) \vec{a} = 12\vec{i} + 3\vec{j} - 4\vec{k}$$

$$\vec{b} = 3\vec{i} + 4\vec{j} + 12\vec{k}$$

$$\vec{a} \cdot \vec{b} = 0 \quad |\vec{a}| = |\vec{b}| = L \quad \Rightarrow \vec{c} = \pm \frac{\vec{a} \times \vec{b}}{L}$$

$$\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} \cdot \vec{b} = 36 + 12 - 48 = 0$$

$$\Rightarrow \vec{a} \perp \vec{b} \quad (1)$$

$$\left. \begin{aligned} |\vec{a}| &= \sqrt{144 + 9 + 16} = \sqrt{169} = 13 \\ |\vec{b}| &= \sqrt{9 + 16 + 144} = \sqrt{169} = 13 \end{aligned} \right\} \Rightarrow |\vec{a}| = |\vec{b}| \quad (2)$$

From (1) and (2) $\Rightarrow \vec{a}$ and $\vec{b}$ are two edges of a cube

$$\vec{c} = \pm \frac{(\vec{a} \times \vec{b})}{L} = \pm \frac{52\vec{i} - 156\vec{j} + 39\vec{k}}{13} = \pm (4\vec{i} - 12\vec{j} + 3\vec{k})$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 12 & 3 & -4 \\ 3 & 4 & 12 \end{vmatrix} = 52\vec{i} - 156\vec{j} + 39\vec{k}$$

$$1.9) \begin{aligned} \vec{v} &= 2\vec{i} - 3\vec{j} - 5\vec{k} & \vec{v} &= \alpha\vec{v}_1 + \beta\vec{v}_2 + \gamma\vec{v}_3 \\ \vec{v}_1 &= 3\vec{i} + \vec{k} & \vec{v} &= 3\alpha\vec{i} + \alpha\vec{k} - \beta\vec{j} - \beta\vec{k} \\ \vec{v}_2 &= -\vec{j} - \vec{k} & & - \gamma\vec{i} + 3\gamma\vec{k} \\ \vec{v}_3 &= -\vec{i} + 3\vec{k} & \vec{v} &= \vec{i}(3\alpha - \gamma) - \beta\vec{j} + \vec{k}(\alpha - \beta + 3\gamma) \end{aligned}$$

$$\begin{cases} 3\alpha - \gamma = 2 \\ -\beta = -3 \\ \alpha - \beta + 3\gamma = -5 \end{cases} \Rightarrow \begin{cases} \beta = 3 \\ 3\alpha - \gamma = 2 \\ \alpha + 3\gamma = -2 \end{cases} \Leftrightarrow \begin{cases} 9\alpha - 3\gamma = 6 \\ \alpha + 3\gamma = -2 \end{cases}$$

$$\stackrel{(*)}{\Rightarrow} \frac{10\alpha = 4}{\gamma = -\frac{4}{5}} \Rightarrow \alpha = \frac{2}{5}$$

$$\Rightarrow \vec{v} = \frac{2}{5}\vec{v}_1 + 3\vec{v}_2 - \frac{4}{5}\vec{v}_3$$

$$1.10) \begin{aligned} \vec{v} &= \vec{i} + \vec{j} + 2\vec{k} \\ \vec{a} &= 2\vec{i} + \vec{j} \\ \vec{b} &= -\vec{i} + 3\vec{k} \\ \vec{c} &= -\vec{i} + \vec{j} - \vec{k} \end{aligned}$$

$$\begin{aligned} \vec{v} &= \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} \\ \vec{v} &= 2\alpha\vec{i} + \alpha\vec{j} - \beta\vec{i} + 3\beta\vec{k} - \gamma\vec{i} + \gamma\vec{j} - \gamma\vec{k} \end{aligned}$$

$$\vec{v} = \vec{i}(2\alpha - \beta - \gamma) + \vec{j}(\alpha + \gamma) + \vec{k}(3\beta - \gamma) = \vec{i} + \vec{j} + 2\vec{k}$$

$$\begin{cases} 2\alpha - \beta - \gamma = 1 \\ \alpha + \gamma = 1 \\ 3\beta - \gamma = 2 \end{cases} \quad [A|B] = \left[\begin{array}{ccc|c} 2 & -1 & -1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & -1 & 2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & -1 & -1 & 1 \\ 0 & 3 & -1 & 2 \end{array} \right] \xrightarrow{L_2 - 2L_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & -1 \\ 0 & 3 & -1 & 2 \end{array} \right] \xrightarrow{L_3 + 3L_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & -1 \\ 0 & 0 & -10 & -1 \end{array} \right] \Leftrightarrow \begin{aligned} \alpha + \gamma &= 1 \\ -\beta - 3\gamma &= 1 \\ \gamma &= -\frac{1}{10}, \alpha = \frac{9}{10}, \beta = \frac{7}{10} \end{aligned}$$

$$\Rightarrow \vec{v} = \frac{9\vec{a} + 7\vec{b} + \vec{c}}{10}$$

$$1.11) \vec{a} = 2\vec{i} + \vec{j}$$

$$\vec{b} = -\vec{i} + 3\vec{k}$$

$$\vec{c} = -\vec{i} + \vec{j} - \vec{k}$$

$\vec{a}, \vec{b}, \vec{c}$ form a basis $\Rightarrow$

$$\begin{vmatrix} 2 & -1 & -1 \\ 1 & 0 & 1 \\ 0 & 3 & -1 \end{vmatrix} \neq 0 = 2 \cdot (-3) - 1 \cdot (-3) = -6 + 3 = -3 \neq 0$$

$$\begin{aligned} \vec{d} &= 2\vec{i} + \vec{j} + \vec{k} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = 2\alpha\vec{i} + \alpha\vec{j} - \beta\vec{i} \\ &+ 3\beta\vec{k} - \gamma\vec{i} + \gamma\vec{j} - \gamma\vec{k} = \vec{i}(2\alpha - \beta - \gamma) + \vec{j}(\alpha + \gamma) \\ &+ \vec{k}(3\beta - \gamma) \Rightarrow \end{aligned}$$

$$\begin{cases} 2\alpha - \beta - \gamma = 2 \\ \alpha + \gamma = 1 \\ 3\beta - \gamma = 1 \end{cases} \Rightarrow [A|B] = \begin{bmatrix} 2 & -1 & -1 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & -1 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & -1 & -1 & 2 \\ 0 & 3 & -1 & 1 \end{bmatrix} \xrightarrow{L_2 - 2L_1} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & 0 \\ 0 & 3 & -1 & 1 \end{bmatrix} \xrightarrow{L_3 + 3L_2}$$

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & -10 & 1 \end{bmatrix} \Rightarrow \begin{cases} \alpha + \gamma = 1 \\ -\beta - 3\gamma = 0 \\ \gamma = -\frac{1}{10}, \alpha = \frac{11}{10}, \beta = \frac{3}{10} \end{cases}$$

$$\Rightarrow \vec{d} = \frac{11\vec{a} + 3\vec{b} - \vec{c}}{10}$$

$$1.12) A(1, -1, 2), B(2, 1, 0), C(3, 2, -6)$$

$$a) \vec{AB} = \vec{i} + 2\vec{j} - 2\vec{k}$$

$$|\vec{AB}| = \sqrt{1 + 4 + 4} = 3$$

$$b) \vec{BC} = \vec{i} + \vec{j} - 6\vec{k}$$

$$\vec{AC} = 2\vec{i} + 4\vec{j} - 8\vec{k}$$

$$4\vec{B} \cdot \vec{BC} = 1 + 2 + 12 = 15$$

$$c) A_{\Delta ABC} = \frac{|\vec{AB} \times \vec{AC}|}{2} = \frac{\sqrt{117}}{2}$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -2 \\ 2 & 3 & -8 \end{vmatrix} = -10\vec{i} + 4\vec{j} - \vec{k}$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{100 + 16 + 1} = \sqrt{117}$$

$$1.13) \vec{u} = 2\vec{i} + 3\vec{j} + 2\vec{k}$$

$$\vec{v} = -\vec{i} + 4\vec{j} + \vec{k}$$

$$\vec{w} = -2\vec{j} + 2\vec{k}$$

$$a) \vec{v} + \vec{w} = \vec{j} + 2\vec{j} + 2\vec{k}$$

$$b) 2\vec{u} - 3\vec{w} = 10\vec{i} + 6\vec{j} - \vec{k}$$

$$c) \vec{u} \cdot \vec{v} = -2 + 12 + 2 = 12$$

$$d) |\vec{v}| = \sqrt{1 + 16 + 1} = 3\sqrt{2}$$

$$e) \vec{u} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 2 \\ 0 & -2 & 1 \end{vmatrix} = 2\vec{i} - 2\vec{j} - 4\vec{k}$$

$$f) \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 2 \\ -1 & 4 & 0 \end{vmatrix} = -8\vec{i} - 2\vec{j} + 11\vec{k}$$

$$|\vec{u} \times \vec{v}| = \sqrt{64 + 4 + 121} = \sqrt{189}$$

$$g) h = \frac{|\vec{u} \times \vec{v}|}{|\vec{v}|} = \frac{\sqrt{189}}{\sqrt{18}}$$

$$h) V = \begin{vmatrix} 2 & 3 & 2 \\ -1 & 4 & 1 \\ 0 & -2 & 2 \end{vmatrix} \cdot \frac{1}{6} = (20 + 6 + 4) \cdot \frac{1}{6} = 5$$

$$1.14) A(0, 1, -1), B(1, 1, 3), C(2, 1, -2), D(3, -1, 2)$$

$$a) \begin{aligned} \vec{AB} &= (1, 0, 4) & V_{ABCD} &= \frac{1}{6} \begin{vmatrix} 1 & 0 & 4 \\ 2 & 0 & -1 \\ 3 & -2 & 3 \end{vmatrix} = \frac{1}{6} (-2 - 16) \\ \vec{AC} &= (2, 0, -1) & &= \frac{18}{6} = 3 \\ \vec{AD} &= (3, -2, 3) \end{aligned}$$

$$b) V_{ABCD} = d(D, ABC) \cdot A_{ABC}$$

$$\begin{aligned} A_{ABC} &= |\vec{AB} \times \vec{AC}| = 9 \\ \vec{AB} \times \vec{AC} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 4 \\ 2 & 0 & -1 \end{vmatrix} = 9\vec{j} \end{aligned}$$

$$d(D, ABC) = \frac{3 V_{ABCD}}{A_{ABC}} = \frac{18}{9} = 2$$

$$c) \angle(\vec{AB}, \vec{CD}) = \alpha \quad \begin{aligned} |\vec{AB}| &= \sqrt{17} \\ |\vec{CD}| &= \sqrt{21} \end{aligned}$$

$$\vec{CD} = \vec{i} - 2\vec{j} + 4\vec{k}$$

$$\vec{AB} \cdot \vec{CD} = 1 + 16 = 17$$

$$\Rightarrow \cos \alpha = \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| \cdot |\vec{CD}|} = \frac{\sqrt{17}}{\sqrt{21}} \Rightarrow \alpha = \arccos\left(\frac{\sqrt{17}}{\sqrt{21}}\right)$$

$$1.15) \vec{a} = 2\vec{i} + 3\vec{j} + (2\lambda - 3)\vec{k}$$

$$\vec{b} = (2+3\lambda)\vec{i} + \lambda\vec{j} - 2\vec{k}$$

$$\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0 \Rightarrow 4 + 6\lambda - 4\lambda + 6 = 0$$

$$2\lambda = -10 \Rightarrow \lambda = -5$$

$$1.16) |\vec{a}| = 4 \quad (3\vec{a} + 5\vec{b}) \perp (\vec{a} - 2\vec{b})$$

$$|\vec{b}| = 2$$

$$(3\vec{a} + 5\vec{b}) \cdot (\vec{a} - 2\vec{b}) = 0$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \alpha = 8 \cos \alpha$$

$$3|\vec{a}|^2 - 6\vec{a} \cdot \vec{b} + 5\vec{a} \cdot \vec{b} - 10|\vec{b}|^2 = 0$$

$$48 - \vec{a} \cdot \vec{b} - 10 = 0$$

$$8 = \vec{a} \cdot \vec{b} \Leftrightarrow 8 \cos \alpha = 8 \Rightarrow \alpha = 0$$

$$1.17) \vec{a} = 6\vec{i} + 2\vec{j} - 3\vec{k}$$

$$\vec{b} = -3\vec{i} + 6\vec{j} - 2\vec{k}$$

$$\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0 \Rightarrow -18 + 12 + 6 = 0 \Leftrightarrow 0 = 0$$

$$|\vec{a}| = \sqrt{36 + 4 + 9} = 7$$

$$|\vec{b}| = \sqrt{9 + 36 + 4} = 7$$

$$\vec{c} = \pm \frac{(\vec{a} \times \vec{b})}{L} \quad \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 2 & -3 \\ -3 & 6 & -2 \end{vmatrix}$$

$$= 14\vec{i} + 21\vec{j} + 42\vec{k} \Rightarrow \vec{c} = \pm (2\vec{i} + 3\vec{j} + 6\vec{k})$$

$$1.18) \vec{a} = (2, 3, 1) \quad \vec{c} = (-2, 1, 2)$$

$$\vec{b} = (1, 1, -2)$$

$$a) \vec{a} \cdot (\vec{b} + \vec{c}) = -2 + 6 + 0 = 4$$

$$\vec{b} + \vec{c} = (-1, 2, 0)$$

$$b) \vec{a} \times (\vec{b} + \vec{c}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & 2 & 0 \end{vmatrix} = -2\vec{i} - \vec{j} + 7\vec{k}$$

$$c) \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 1 & -2 \\ -2 & 1 & 2 \end{vmatrix} = 8 + 6 + 3 = 17$$

$$d) \vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{a} \cdot \vec{c}) - \vec{c} \cdot (\vec{a} \cdot \vec{b})$$

$$\vec{a} \cdot \vec{c} = -4 + 3 + 2 = 1$$

$$\vec{a} \cdot \vec{b} = 2 + 3 - 2 = 3$$

$$\Rightarrow \vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} - 3\vec{c} = (7, -2, -8)$$

$$1. 19) \vec{DA} \cdot \vec{BC} + \vec{DB} \cdot \vec{CA} + \vec{DC} \cdot \vec{AB} = 0$$

$$\vec{DA} = \vec{a} - \vec{d} \quad \vec{DB} = \vec{b} - \vec{d} \quad \vec{DC} = \vec{c} - \vec{d}$$

$$\vec{BC} = \vec{c} - \vec{b} \quad \vec{CA} = \vec{a} - \vec{c} \quad \vec{AB} = \vec{b} - \vec{a}$$

$$\vec{DA} \cdot \vec{BC} = \vec{a} \cdot \vec{c} - \vec{a} \cdot \vec{b} - \vec{d} \cdot \vec{c} + \vec{b} \cdot \vec{d}$$

$$\vec{DB} \cdot \vec{CA} = \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{a} \cdot \vec{d} + \vec{d} \cdot \vec{c}$$

$$\vec{DC} \cdot \vec{AB} = \vec{b} \cdot \vec{c} - \vec{a} \cdot \vec{c} - \vec{d} \cdot \vec{b} + \vec{a} \cdot \vec{d}$$

$$\begin{aligned} (+) & \underline{\hspace{10em}} \\ &= \vec{a} \cdot \vec{c} - \vec{a} \cdot \vec{d} - \vec{d} \cdot \vec{c} + \vec{b} \cdot \vec{d} + \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{a} \cdot \vec{d} + \vec{d} \cdot \vec{c} + \vec{b} \cdot \vec{c} - \vec{a} \cdot \vec{c} - \vec{d} \cdot \vec{b} + \vec{a} \cdot \vec{d} \\ &= 0 \end{aligned}$$

$$\left. \begin{aligned} \text{b) if } \vec{DA} \perp \vec{BC} &\Rightarrow \vec{OA} \cdot \vec{BC} = 0 \quad (1) \\ \text{if } \vec{DB} \perp \vec{CA} &\Rightarrow \vec{OB} \cdot \vec{CA} = 0 \quad (2) \\ \vec{OA} \cdot \vec{BC} + \vec{OB} \cdot \vec{CA} + \vec{OC} \cdot \vec{AB} &= 0 \quad (3) \end{aligned} \right\} \Rightarrow \vec{OC} \cdot \vec{AB} = 0$$

$$\Rightarrow \vec{OC} \perp \vec{AB}$$

$$1.20) \quad \vec{g} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

$$\text{a) } \vec{AG} + \vec{BG} + \vec{CG} = \vec{0}$$

$$\vec{g} - \vec{a} + \vec{g} - \vec{b} + \vec{g} - \vec{c} = \vec{0}$$

$$3\vec{g} - (\vec{a} + \vec{b} + \vec{c}) = \vec{0}$$

$$\vec{a} + \vec{b} + \vec{c} - (\vec{a} + \vec{b} + \vec{c}) = \vec{0} \Rightarrow \vec{0} = \vec{0}$$

$$\text{b) } 3\vec{MG} = \vec{MA} + \vec{MB} + \vec{MC}$$

$$3\vec{g} - 3\vec{m} = \vec{a} + \vec{b} + \vec{c} - 3\vec{m}$$

$$\vec{0} = \vec{0}$$

$$1.21) \quad |\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 \cdot |\vec{b}|^2$$

(Lagrange identity)

$$|\vec{a}|^2 \cdot |\vec{b}|^2 \cdot \sin^2 \theta + |\vec{a}|^2 \cdot |\vec{b}|^2 \cdot \cos^2 \theta = |\vec{a}|^2 \cdot |\vec{b}|^2$$

$$\Rightarrow |\vec{a}|^2 \cdot |\vec{b}|^2 (\sin^2 \theta + \cos^2 \theta) = |\vec{a}|^2 \cdot |\vec{b}|^2$$

$$\Rightarrow |\vec{a}|^2 \cdot |\vec{b}|^2 = |\vec{a}|^2 \cdot |\vec{b}|^2 \quad \textcircled{A}$$

$$1.22) \quad |\vec{w}| = 3$$

$$W = (x, y, 0)$$

$$\vec{w} \perp \vec{Oz}$$

$$\angle(\vec{Ox}, \vec{w}) = 45^\circ$$

$$\vec{Ox} = (1, 0, 0)$$

$$O_x \cdot W = X \Rightarrow O_x \cdot W = |W| \cdot |O_x| \cdot \cos 45^\circ$$

$$X = 3 \cdot \cos 45^\circ = \frac{3\sqrt{2}}{2}$$

$$|\vec{W}| = \sqrt{\frac{18}{1} + y^2} \Rightarrow 9 = \frac{9}{2} + y^2 \Rightarrow y^2 = \frac{9}{2} \Rightarrow y = \pm \frac{3\sqrt{2}}{2}$$

$$\Rightarrow \vec{W} = \frac{3\sqrt{2}}{2} (\vec{i} \pm \vec{j})$$

$$1.23) \vec{V} = -\frac{\sqrt{3}}{2} \vec{i} + \frac{1}{2} \vec{j} \quad \vec{V} \cdot O_x = -\frac{\sqrt{3}}{2}$$

$$O_x = (1, 0, 0)$$

$$\cos \alpha = \frac{\vec{V} \cdot O_x}{|\vec{V}| \cdot |O_x|} = \frac{-\frac{\sqrt{3}}{2}}{1} = -\frac{\sqrt{3}}{2}$$

$$|\vec{V}| = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$$

$$\Rightarrow \alpha = \frac{5\pi}{6}$$

$$1.24) \vec{V} = 3\vec{i} - \vec{j} + 2\vec{k}$$

$$\vec{u} = \vec{j} - 2\vec{k}$$

$$\vec{V} \times \vec{u} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -1 & 2 \\ 0 & 1 & -2 \end{vmatrix} = 0\vec{i} + 6\vec{j} + 3\vec{k}$$

$$|\vec{V} \times \vec{u}| = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$$

$$\Rightarrow |\vec{V}| = \sqrt{9 + 1 + 4} = \sqrt{14}$$

$$\Rightarrow h = \frac{|\vec{V} \times \vec{u}|}{|\vec{V}|} = \sqrt{\frac{45}{14}}$$

$$1.25) \vec{a} = 3\vec{i} - \vec{j} + 2\vec{k}$$

$$\vec{b} = \vec{j} - 2\vec{k}$$

$$\vec{c} = \vec{j} + 4\vec{k}$$

$$V = \vec{a} \cdot (\vec{b} \times \vec{c}) =$$

$$\begin{vmatrix} 3 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 1 & 4 \end{vmatrix} = 18$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & 0 \\ 2 & -1 & 1 \end{vmatrix} = 6\vec{j} + 3\vec{k} \Rightarrow |\vec{a} \times \vec{b}| =$$

$$\sqrt{45} \Rightarrow h = \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{a} \times \vec{b}|} = \frac{12}{\sqrt{45}} = \frac{6\sqrt{5}}{5}$$

$$\begin{aligned} 1.26) \quad \vec{a} &= 3\vec{i} - \vec{j} + \alpha\vec{k} & \vec{a} \times (\vec{b} \times \vec{c}) &= \\ \vec{b} &= \vec{j} + 2\vec{k} & \vec{b} \cdot (\vec{a} \cdot \vec{c}) - \vec{c} \cdot (\vec{a} \cdot \vec{b}) &= \\ \vec{c} &= 3\vec{i} - \vec{k} & \vec{a} \cdot \vec{c} &= 9 - \alpha \\ & & \vec{a} \cdot \vec{b} &= -1 + 2\alpha \end{aligned}$$

$$\vec{a} \times (\vec{b} \times \vec{c}) \parallel yOz$$

$$\Rightarrow x=0 \Rightarrow \vec{i}(-3 + 6\alpha) = 0 \Rightarrow \alpha = \frac{1}{2}$$

$$\begin{aligned} 1.27) \quad \vec{a} &= \vec{i} + \vec{j} + 2\vec{k} & \vec{a} \times (\vec{b} \times \vec{c}) &= \vec{b} \cdot (\vec{a} \cdot \vec{c}) - \\ \vec{b} &= 2\vec{i} - \vec{j} + \lambda\vec{k} & \vec{c} \cdot (\vec{a} \cdot \vec{b}) &= \\ \vec{c} &= \vec{i} - 2\vec{j} + \vec{k} & \vec{a} \cdot \vec{c} &= 1 - 2 + 2 = 1 \\ & & \vec{a} \cdot \vec{b} &= 2 - 1 + 2\lambda = 1 + 2\lambda \end{aligned}$$

$$\vec{a} \times (\vec{b} \times \vec{c}) \parallel xOy \Rightarrow z=0$$

$$\lambda - 1 - 2\lambda = 0 \Rightarrow \lambda = -1$$

$$\begin{aligned} 1.28) \quad \vec{a} &= \vec{i} - m\vec{j} + 3\vec{k} & \vec{a}, \vec{b}, \vec{c} \text{ coplanar} &\Leftrightarrow \\ \vec{b} &= m\vec{i} - \vec{j} + \vec{k} & \begin{vmatrix} 1 & -m & 3 \\ m & -1 & 1 \\ 3 & 1 & -1 \end{vmatrix} &= 0 \\ \vec{c} &= 3\vec{i} + \vec{j} - \vec{k} & & \end{aligned}$$

$$\begin{aligned} \Leftrightarrow \begin{vmatrix} 1 & -m & 3 \\ 0 & m^2-1 & 1-3m \\ 0 & 3m+1 & -10 \end{vmatrix} &= 0 \quad \Leftrightarrow 10 - 10m^2 + 9m^2 - 1 = 0 \\ L_2 - mL_1 & \\ L_3 - 3L_1 & \\ \Rightarrow -m^2 + 9 = 0 \Rightarrow m = \pm 3 \end{aligned}$$

$$\text{For } m=1 \Rightarrow V = \begin{vmatrix} 1 & -1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -1 \end{vmatrix} = |-1+9| = 8$$

$$1.32) \quad \vec{a} \times (\vec{b} \times \vec{c}) + \vec{c} \times (\vec{a} \times \vec{b}) + \vec{b} \times (\vec{c} \times \vec{a}) = \vec{0}$$

(Jacobi's identity)

$$\begin{aligned} & \vec{b} \cdot (\vec{a} \times \vec{c}) - \vec{c} \cdot (\vec{a} \times \vec{b}) + \vec{a} \cdot (\vec{c} \times \vec{b}) - \vec{b} \cdot (\vec{c} \times \vec{a}) \\ & + \vec{c} \cdot (\vec{b} \times \vec{a}) - \vec{a} \cdot (\vec{b} \times \vec{c}) = 0 \end{aligned}$$

$$1.31) \quad \vec{a} = \vec{m} + 2\vec{n} \quad |\vec{m}| = 5 \quad \angle(\vec{m}, \vec{n}) = \frac{\pi}{2}$$

$$\vec{b} = \vec{m} - 3\vec{n} \quad |\vec{m}| = 3$$

$$a) \quad \vec{d}_1 = \vec{a} + \vec{b} = 2\vec{m} - \vec{n} \quad |\vec{d}_1|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$$

$$\vec{d}_2 = \vec{a} - \vec{b} = 5\vec{n} \quad = (2\vec{m} - \vec{n}) \cdot (2\vec{m} - \vec{n})$$

$$= 4|\vec{m}|^2 - 4\vec{m} \cdot \vec{n} + |\vec{n}|^2 = 4 \cdot 25 + 9 = 109 \Rightarrow |\vec{d}_1| = \sqrt{109}$$

$$\vec{m} \cdot \vec{n} = 5 \cdot 3 \cdot \cos \frac{\pi}{2} = 0$$

$$|\vec{d}_2|^2 = 5\vec{n} \cdot 5\vec{n} = 25 \cdot 9 = 225 \Rightarrow |\vec{d}_2| = 15$$

$$b) \quad \vec{d}_1 \cdot \vec{d}_2 = |\vec{d}_1| \cdot |\vec{d}_2| \cdot \cos \theta \Rightarrow \cos \theta = \frac{-45}{15 \cdot \sqrt{109}} = -\frac{3}{\sqrt{109}}$$

$$\vec{d}_1 \cdot \vec{d}_2 = 10\vec{m} \cdot \vec{n} - 5|\vec{n}|^2 = -45 \Rightarrow \theta = \arccos\left(-\frac{3}{\sqrt{109}}\right)$$

$$\begin{aligned}
 c) \quad \vec{a} \times \vec{b} &= (\vec{m} + 2\vec{n}) \times (\vec{m} - 3\vec{n}) = \\
 &= -3(\vec{m} \times \vec{n}) + 2(\vec{n} \times \vec{m}) = -5(\vec{m} \times \vec{n}) \\
 &= |\vec{a} \times \vec{b}| = 5|\vec{m} \times \vec{n}| = 5 \cdot 5 \cdot 3 \cdot \sin \frac{\pi}{2} = 15 \cdot 5 = 75
 \end{aligned}$$

$$\begin{aligned}
 1.30) \quad \vec{a} &= 3\vec{a} + 2\vec{b} & |\vec{a}| &= 2 & \angle(\vec{a}, \vec{b}) &= \frac{\pi}{3} \\
 \vec{w} &= 2\vec{a} - \vec{b} & |\vec{b}| &= 3
 \end{aligned}$$

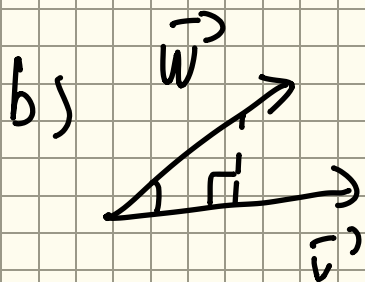
$$a) \quad \vec{a} \cdot \vec{b} = 2 \cdot 3 \cdot \frac{1}{2} = 3$$

$$\vec{u} \cdot \vec{w} = 6|\vec{a}|^2 - 2|\vec{b}|^2 + \vec{a} \cdot \vec{b} = 24 - 18 + 3 = 9$$

$$|\vec{u}|^2 = 9|\vec{a}|^2 + 4|\vec{b}|^2 + 12\vec{a} \cdot \vec{b} = 36 + 36 + 36 = 108$$

$$|\vec{w}|^2 = 4|\vec{a}|^2 + |\vec{b}|^2 - 4\vec{a} \cdot \vec{b} = 16 + 9 - 12 = 13$$

$$\begin{aligned}
 \Rightarrow |\vec{u}| &= \sqrt{108} & |\vec{v}| &= \sqrt{13} \\
 & & \Rightarrow \cos \theta &= \frac{9}{\sqrt{108} \cdot \sqrt{13}}
 \end{aligned}$$



$$\cos \alpha = \frac{p\vec{r}}{|\vec{w}|} \Rightarrow p\vec{r} = \vec{w} \cdot \cos \alpha$$

$$(\Rightarrow) p\vec{r} = \frac{|\vec{u} \cdot \vec{w}|}{|\vec{u}|} = \frac{9}{6\sqrt{3}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$c) \quad |\vec{u} \times \vec{v}| = |(3\vec{a} + 2\vec{b}) \times (2\vec{a} - \vec{b})| \dots$$

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1.2c) For pos $\vec{a}, \vec{b}, \vec{c}, \vec{h}$ the altitude coord
are $(\vec{h} - \vec{a}) \cdot (\vec{c} - \vec{b}) = 0$ and $(\vec{h} - \vec{b}) \cdot (\vec{a} - \vec{c}) = 0$

$$AH \perp BC \Rightarrow (\vec{h} - \vec{a}) \cdot (\vec{c} - \vec{b}) = 0$$

$$BH \perp AC \Rightarrow (\vec{h} - \vec{b}) \cdot (\vec{a} - \vec{c}) = 0$$

Se adunăm și se rezolvă